

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 07

Sequential Circuit



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Recap of Previous Lecture



- Asynch. Counter
-
-



Topics to be Covered

Synchronous Counter

[Question]



We have 10-bit ripple counter which is counting in up sequence. Two most significant bits are NAND together to generate an output y . Output y is periodic waveform with T_{ON} (time duration for which y is high in one period) $15000 \mu\text{sec}$. Then the clock frequency (kHz) 51.2.

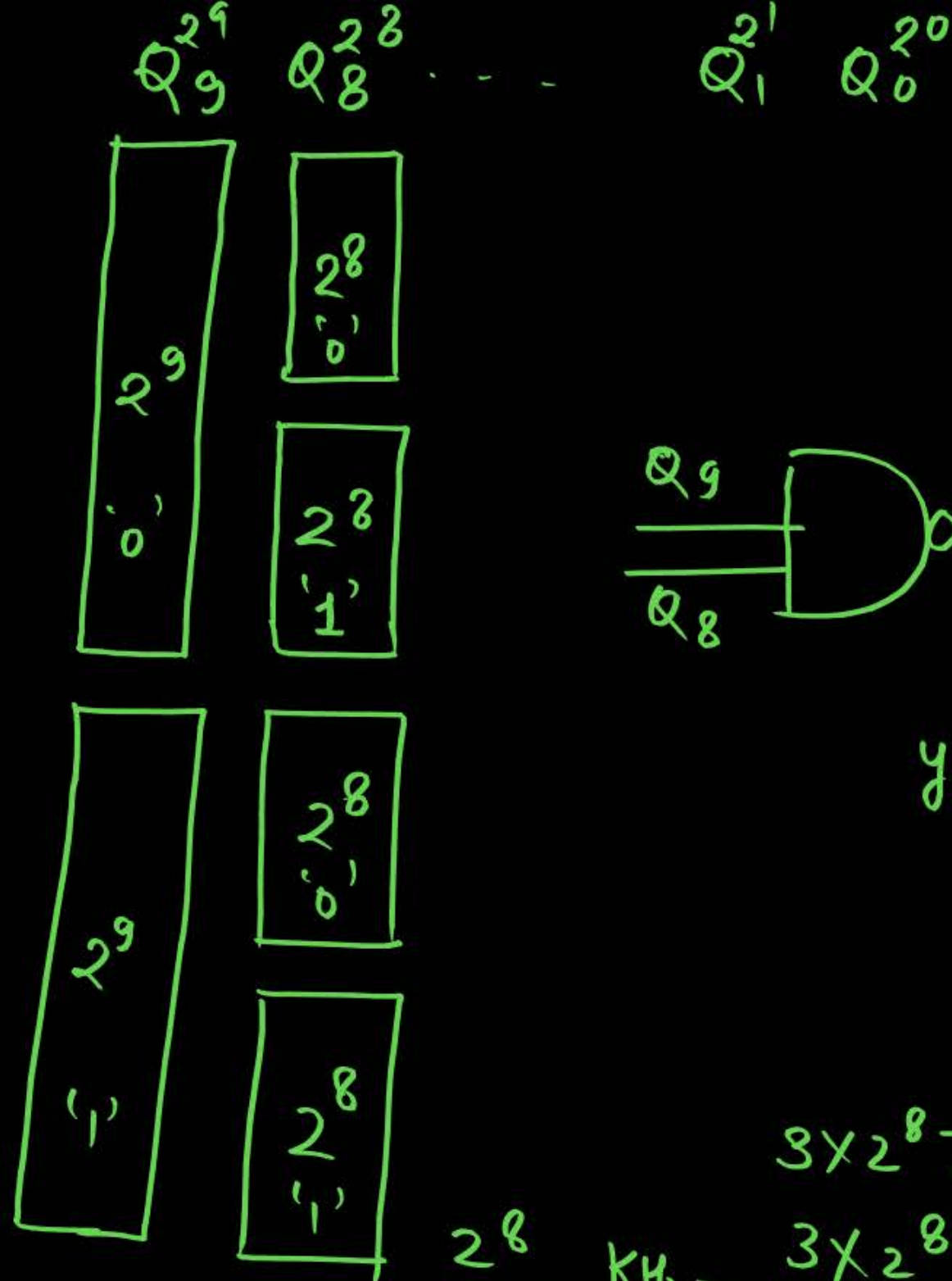
✓ 0 - 1023 $\rightarrow$ MOD no. = 1024

$Q_9 Q_8$

Q_0



$$f_{\text{CLK}} = \frac{256}{5} \text{ KHz} = 51.2 \text{ KHz}$$



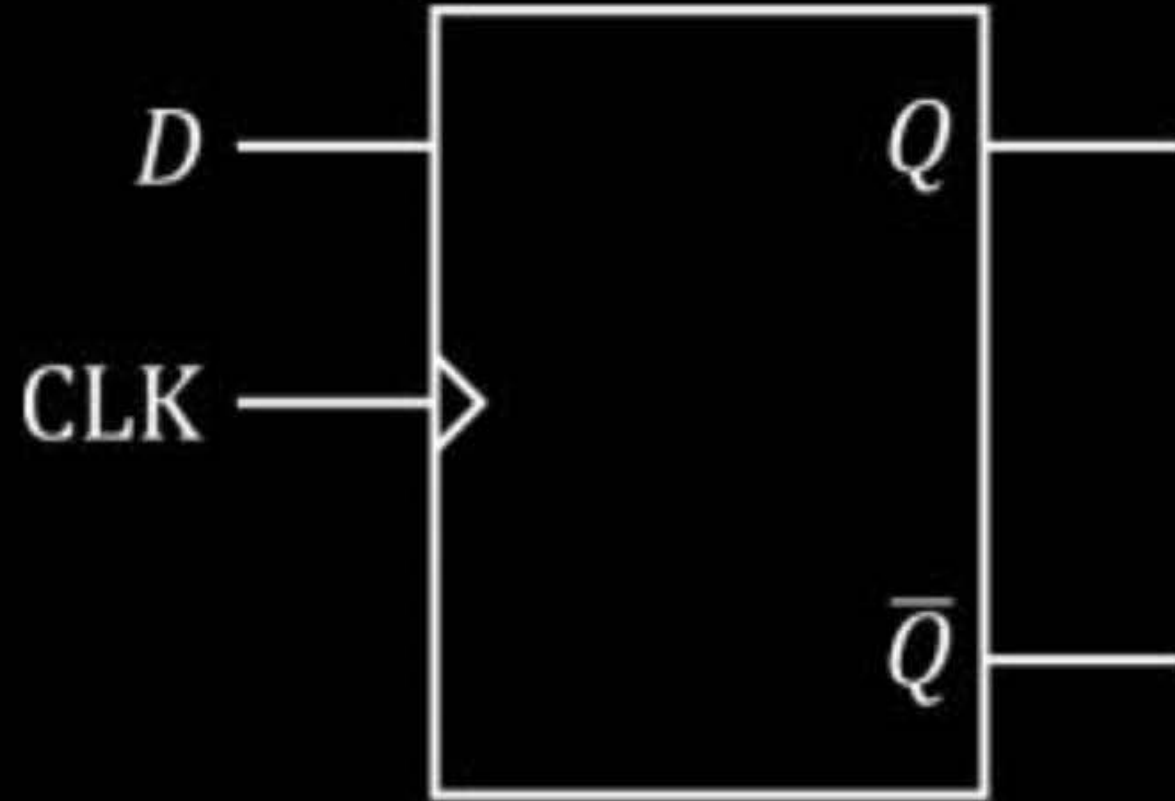
$$\frac{2^8}{5} \text{ KHz} = \frac{3 \times 2^8}{15000 \times 10^{-6} \text{ sec}} = f_{\text{CLK}}$$

[D-FF and T-FF]

$$Q(n+1) = D$$



- What is D-FF ?



- Characteristic table :

D	$Q(n)$	$Q(n+1)$
0	0	0
0	1	0
1	0	1
1	1	1

$$Q(n+1)[D, Q(n)] = \Sigma(2, 3)$$

 $\Rightarrow$

$Q(n+1)$

	$\overline{Q(n)}$	$Q(n)$
$\overline{D}$		
D	1	1

$$Q(n+1) = D$$

- Excitation table :

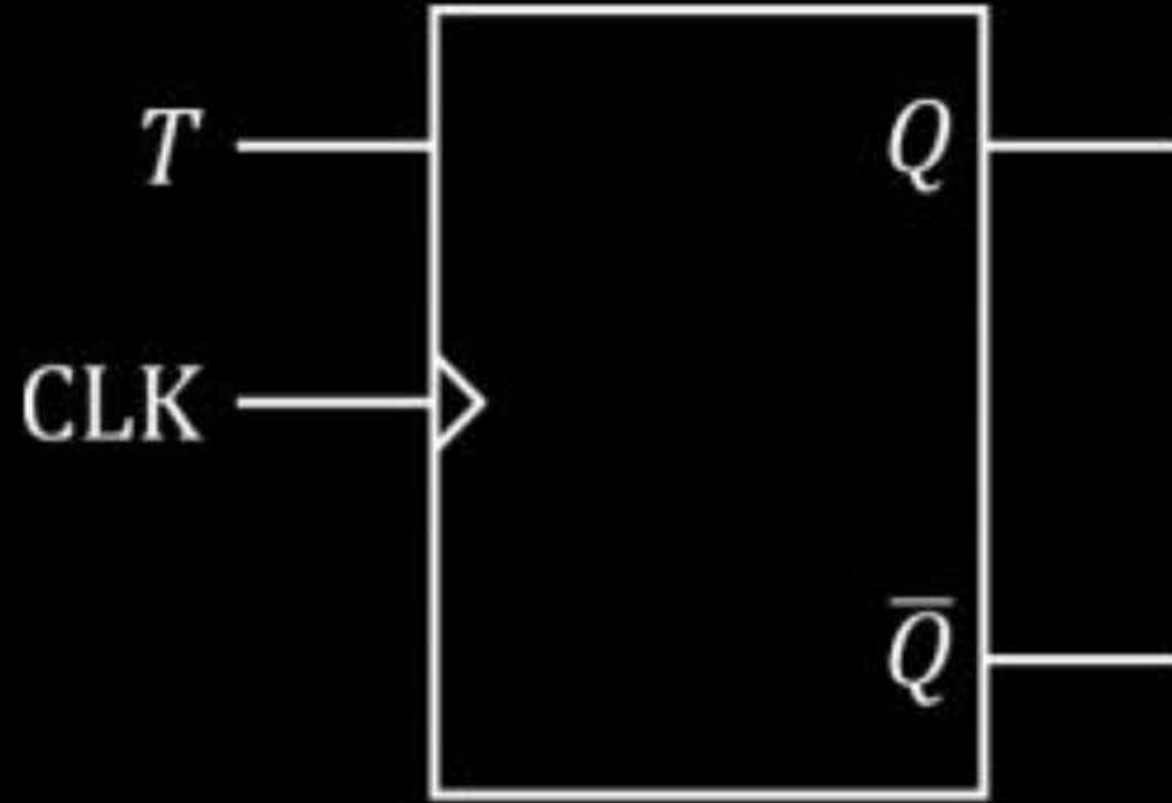
$Q(n)$	$Q(n+1)$	D
0	0	0
0	1	1
1	0	0
1	1	1

- IMP Points :

- Level triggered D-flip is also called transparent latch.

[T-FF]

- What is T-FF?



$$Q(n+1) = T \oplus Q(n)$$

$$T=1 \longrightarrow Q(n+1) = 1 \oplus Q(n)$$
$$Q(n+1) = \overline{Q(n)}$$

$$T=0 \longrightarrow Q(n+1) = 0 \oplus Q(n)$$
$$Q(n+1) = Q(n)$$

- Characteristic table :

T	$Q(n)$	$Q(n+1)$
0	0	0
0	1	1
1	0	1
1	1	0

 $\Rightarrow$

	$\overline{Q(n)}$	$Q(n)$
$\bar{T}$		1
T	1	

$$\begin{aligned}
 Q(n+1)[T, Q(n)] &= \Sigma 1, 2 \\
 &= \bar{T}Q(n) + T\overline{Q(n)} \\
 Q(n+1) &= T \oplus Q(n)
 \end{aligned}$$

- Excitation table :

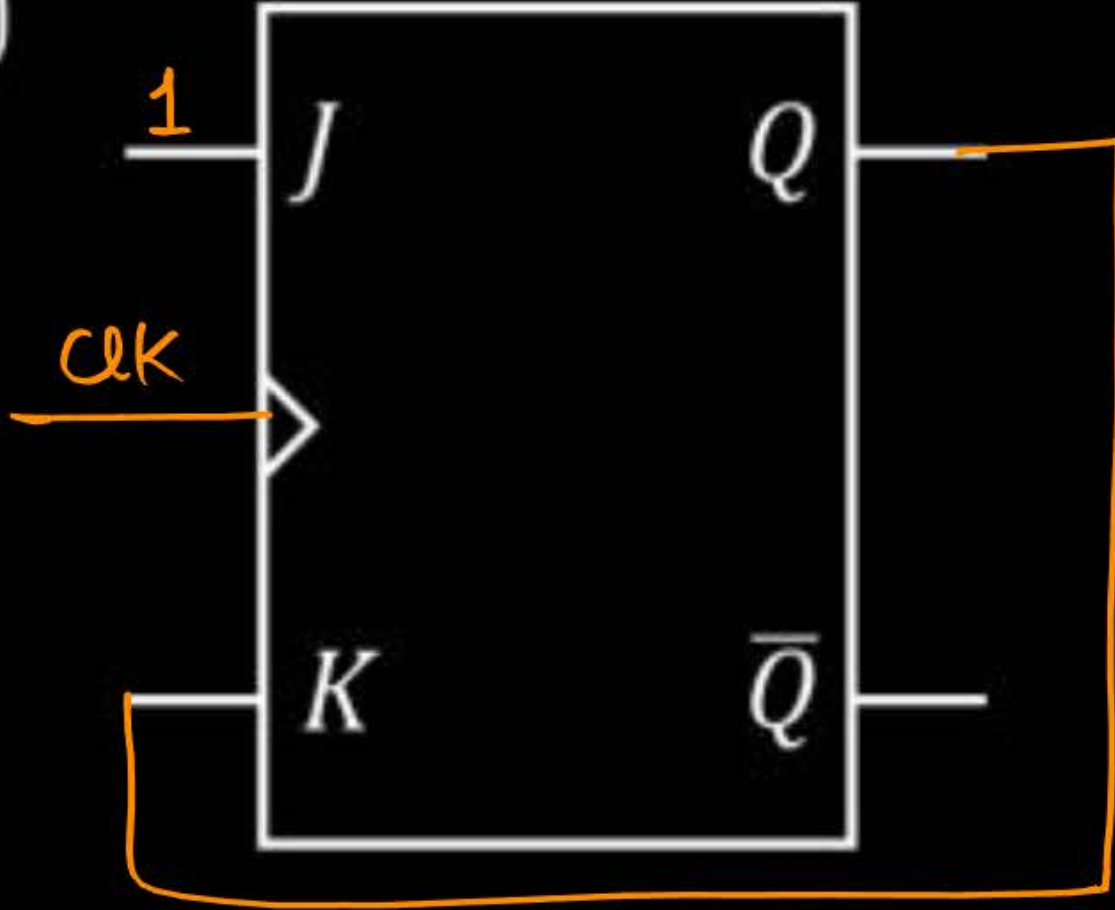
$Q(n)$	$Q(n+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

[Toggle mode of operation]

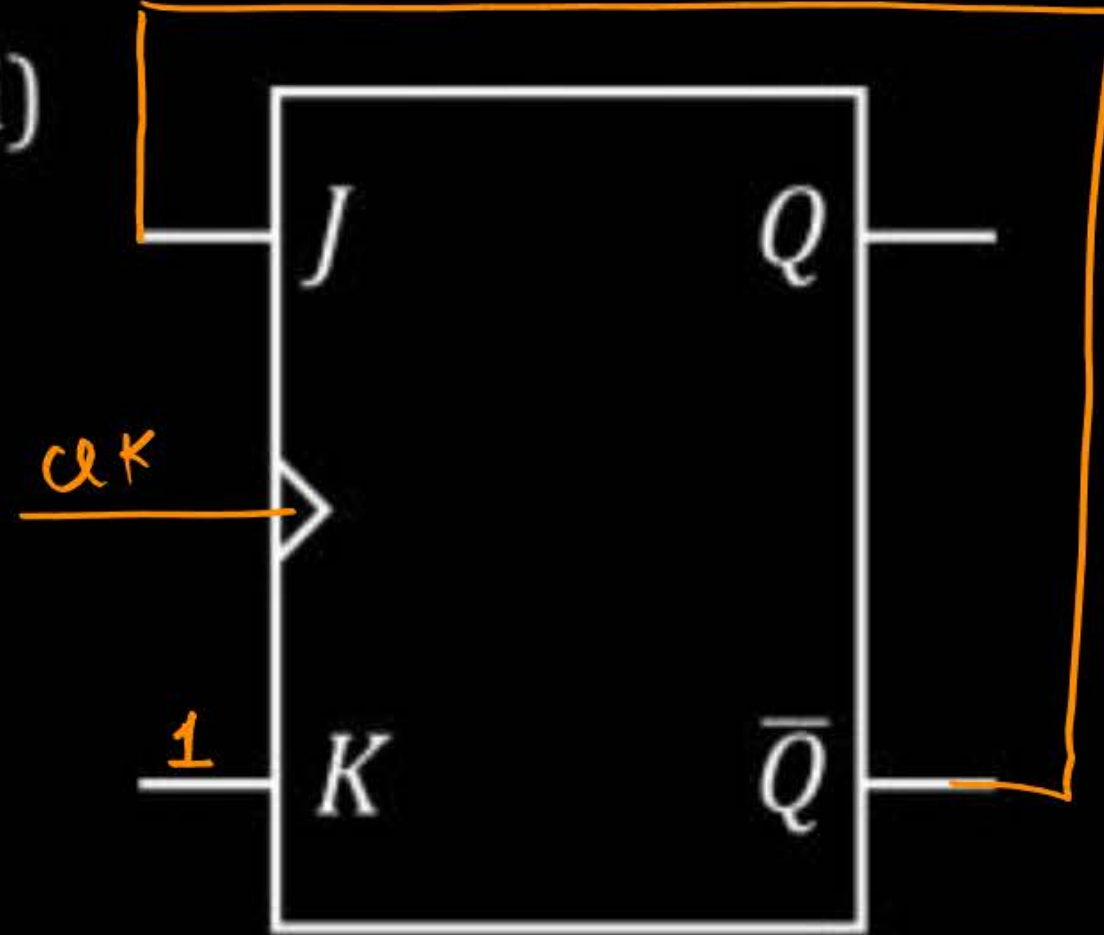
$$Q(n+1) = \bar{Q}(n)$$



(i)



(ii)



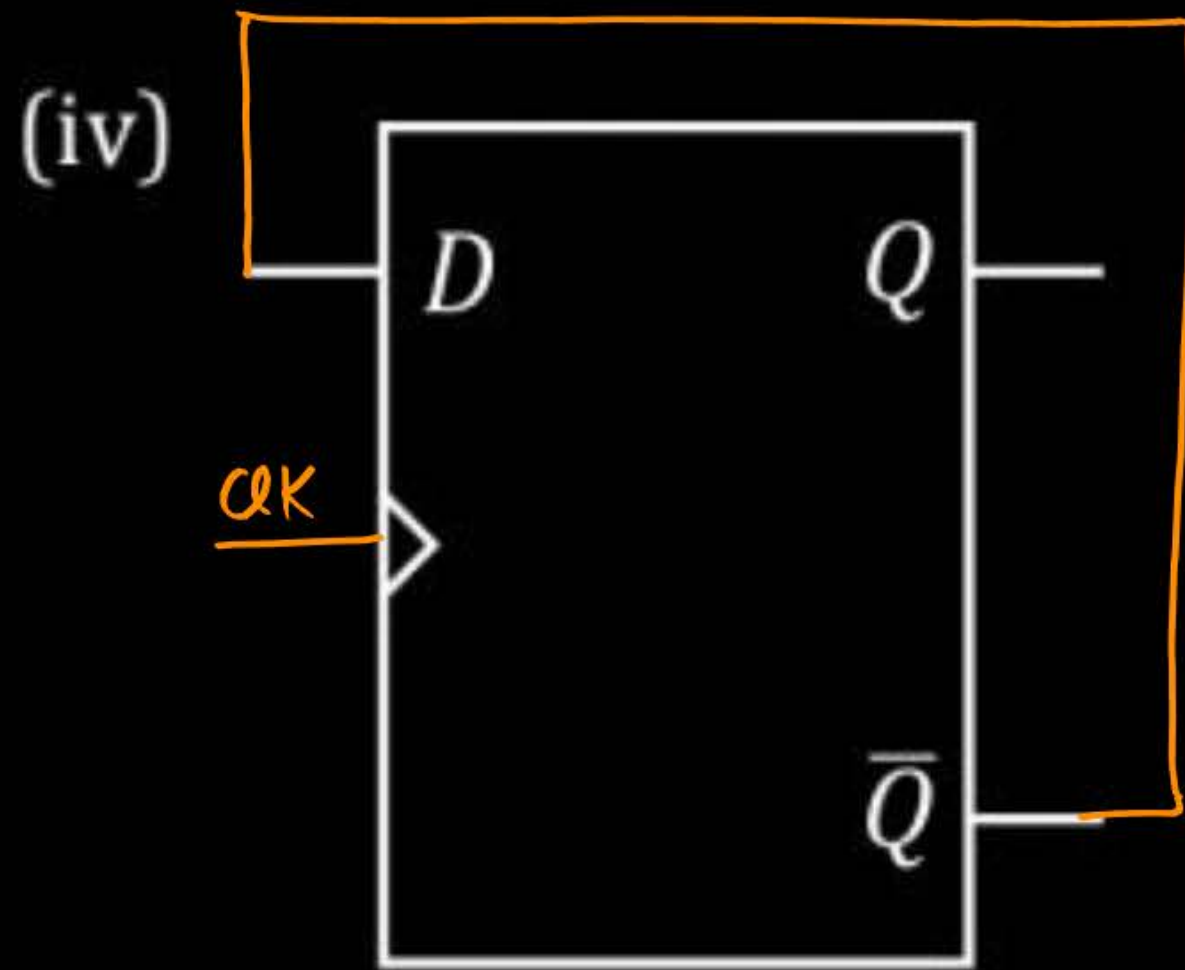
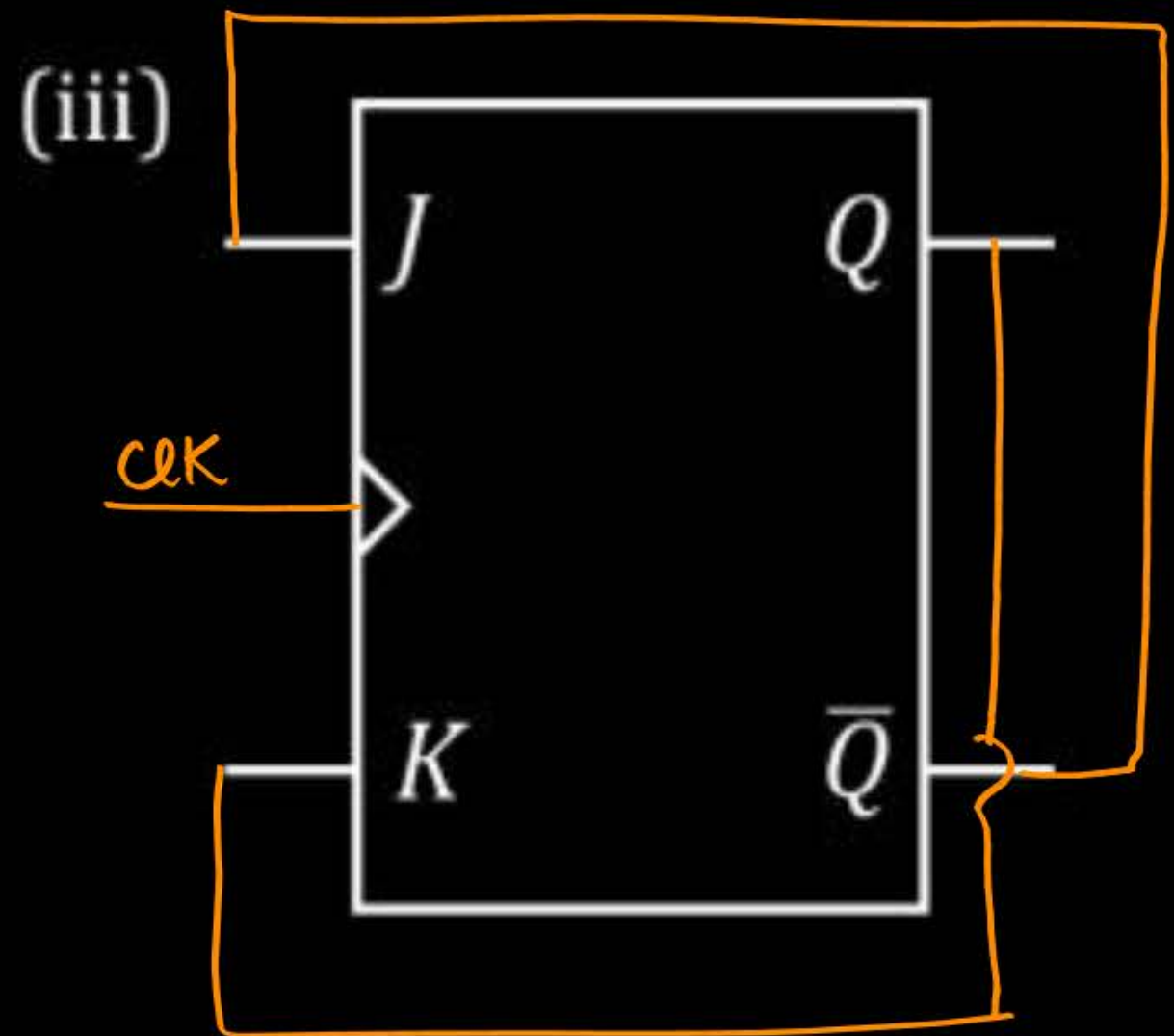
$$Q(n+1) = J \bar{Q}(n) + \bar{K} Q(n)$$

$$Q(n+1) = 1 \cdot \bar{Q}(n) + 0 \cdot Q(n)$$

$$Q(n+1) = \bar{Q}(n)$$

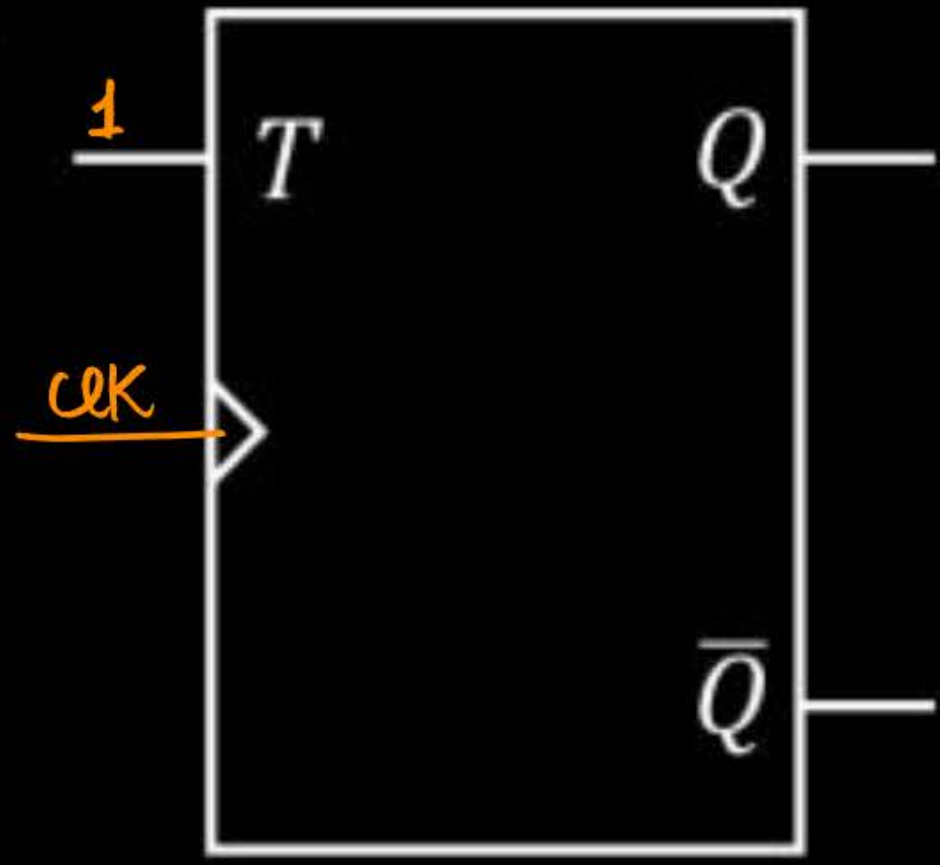
$$J \begin{cases} 1 \\ \bar{Q}(n) \end{cases}$$

$$K \begin{cases} 1 \\ Q(n) \end{cases}$$



$$Q(n+1) = D = \bar{Q}(n)$$

(v)



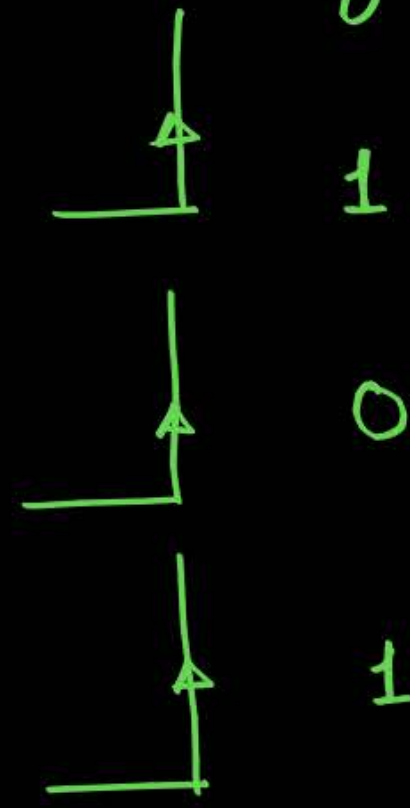
$$Q(n+1) = T \oplus Q(n)$$

$$Q(n+1) = 1 \oplus Q(n)$$

$$Q(n+1) = \overline{Q}(n)$$

$$Q(n+1) = \overline{Q}(n)$$

$f_{clk}/2$
Q



MOD-2 counter

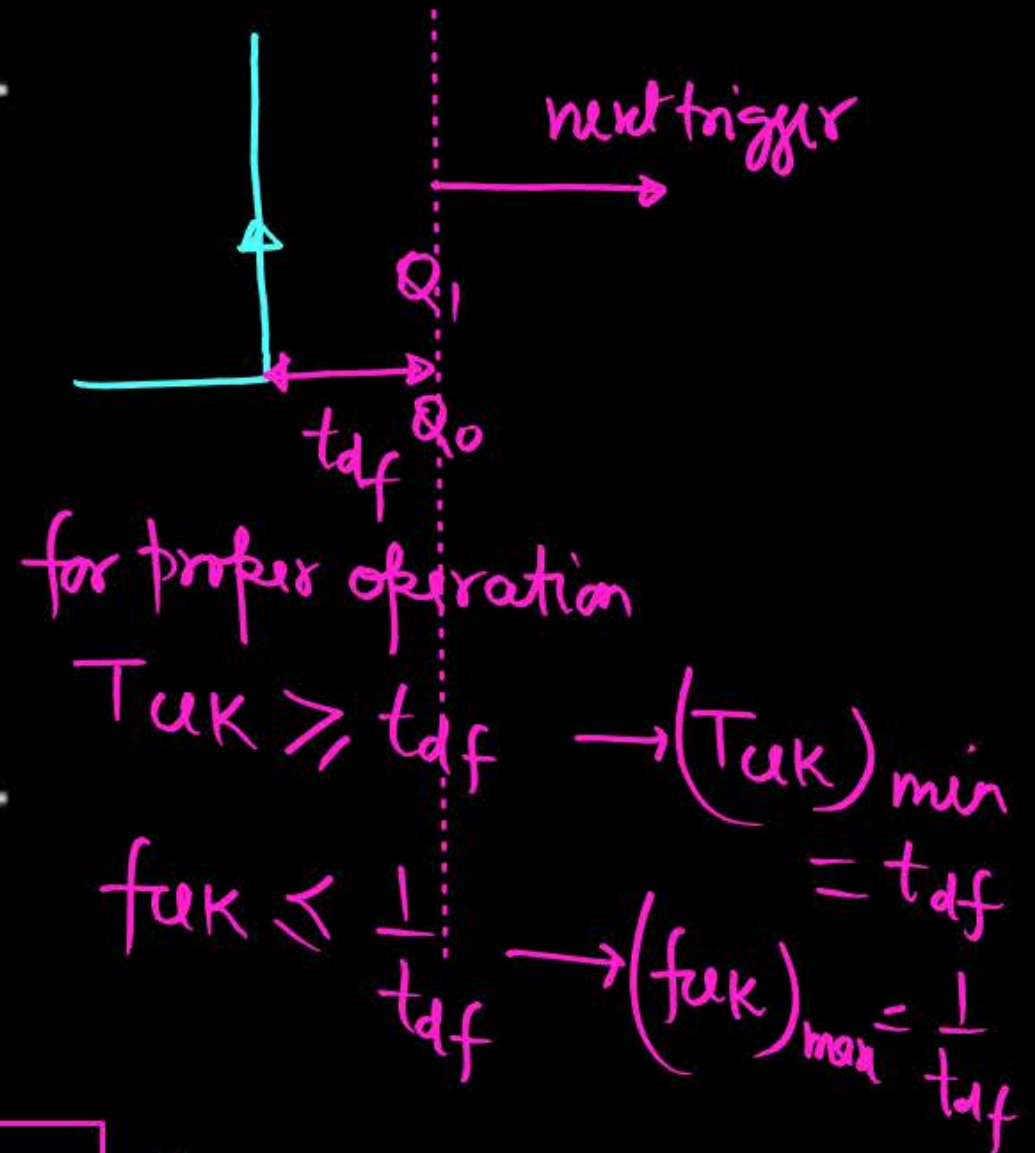
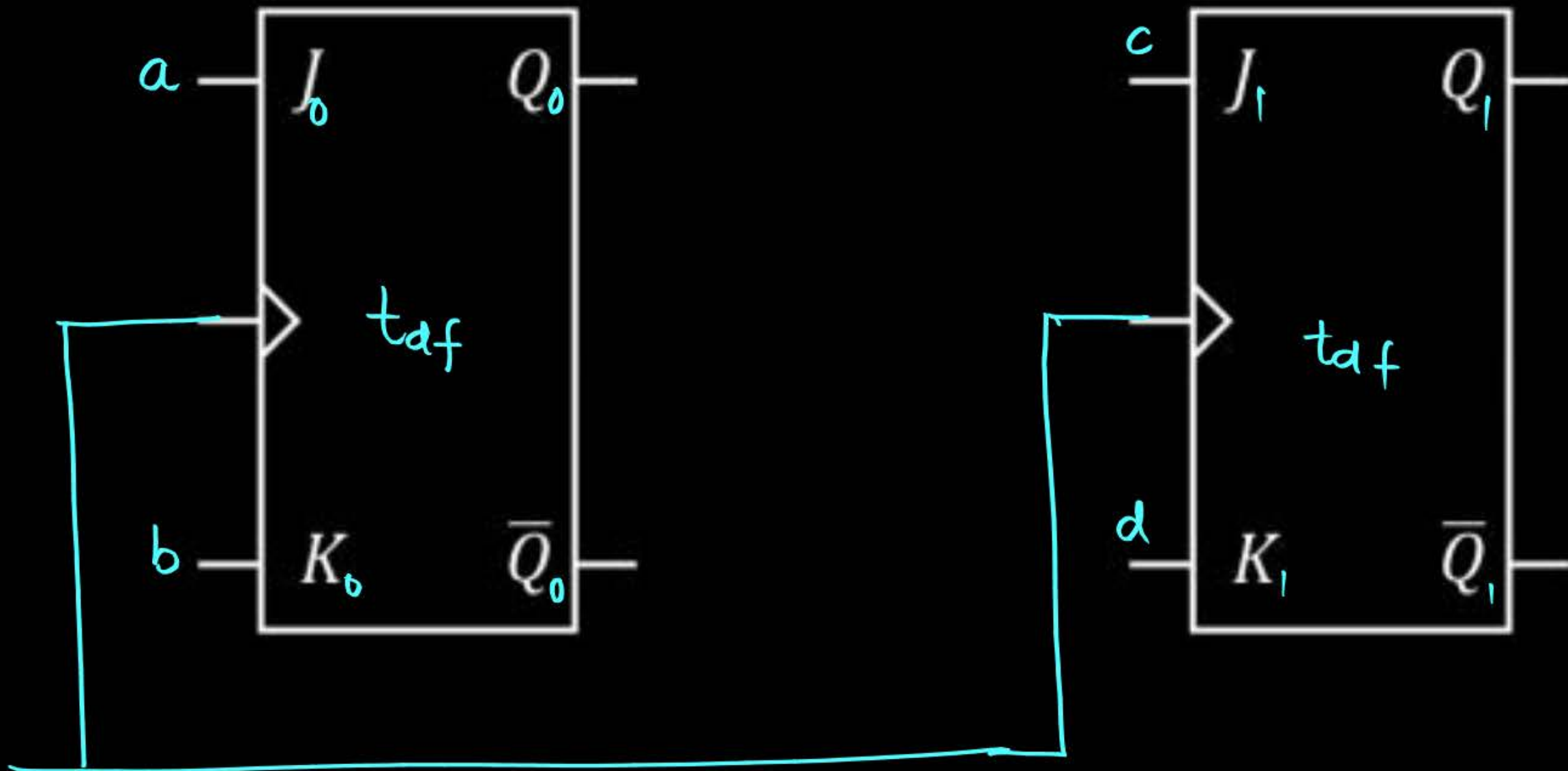
[Synchronous Counter]



- What is synchronous counter ?

↳ All the ffs are driven by same external clock.

- Lets take a synchronous counter and analyze it :



Decoding Error does not occur in synchronous counter. ✓✓

How to design a Synchronous Counter

- 0 - 2 - 3 - 1 $\rightarrow$ MODno = 4

Design above counter using JK-FF $J_1 [Q_1, Q_0] = \Sigma$

Q_1	Q_0	Q_1^+	Q_0^+	J_1	K_1	J_0	K_0
0	0	1	0	1	X	0	X
0	1	0	0	0	X	X	1
1	0	1	1	X	0	1	X
1	1	0	1	X	1	X	0

$$J_1[Q_1, Q_0] = \Sigma 0 + d\Sigma(2, 3)$$

$$K_1[Q_1, Q_0] = \Sigma 3 + d\Sigma(0, 1)$$

$$J_0[Q_1, Q_0] = \Sigma 2 + d\Sigma(1, 3)$$

$$K_0[Q_1, Q_0] = \Sigma 1 + d\Sigma(0, 2)$$

J_1 \ $\overline{Q_0}$ Q_0

$\overline{Q_1}$	1	
Q_1	X	X

$$J_1 = \overline{Q_0}(n)$$

K_1 \ $\overline{Q_0}$ Q_0

$\overline{Q_1}$	X	X
Q_1		1

$$K_1 = Q_0(n)$$

J_0 \ $\overline{Q_0}$ Q_0

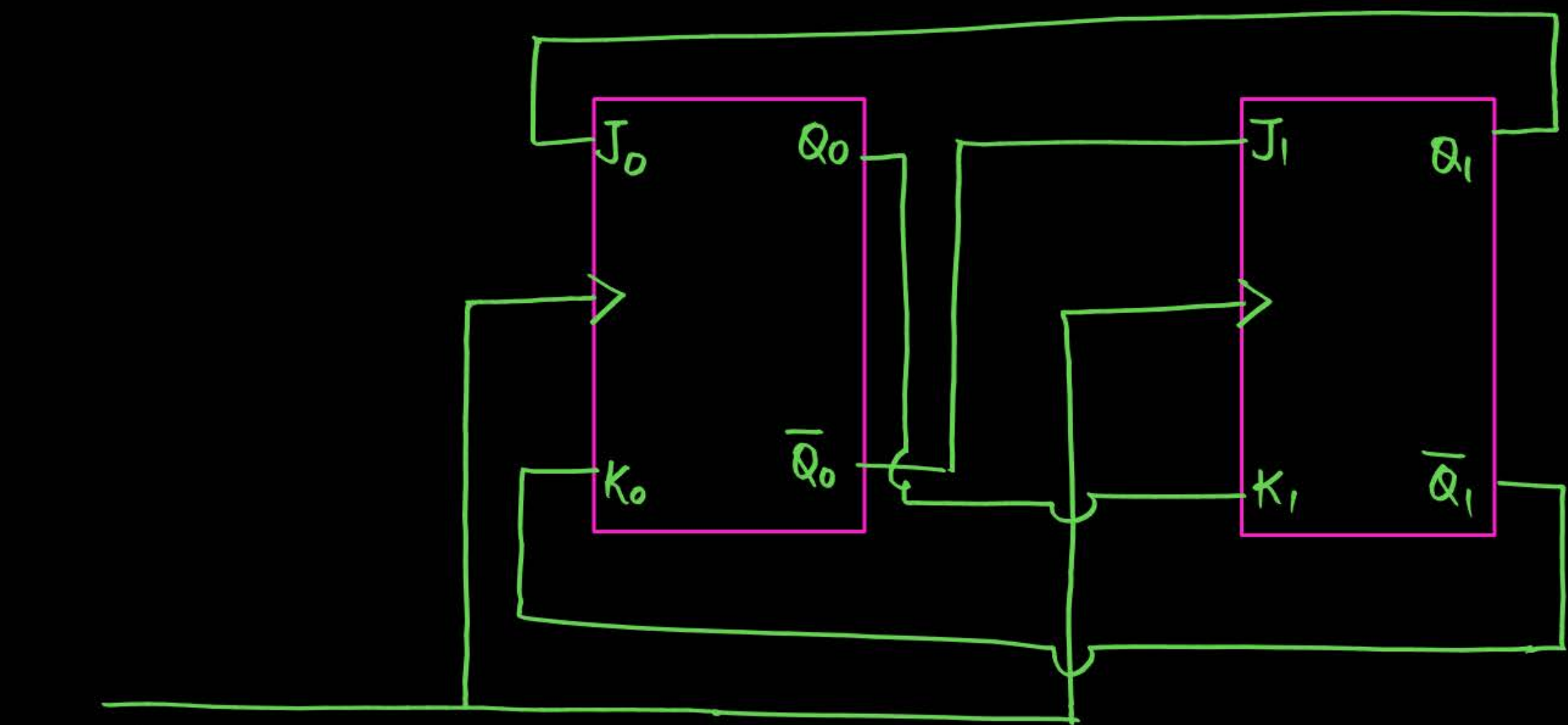
$\overline{Q_1}$		X
Q_1	1	X

$$\overline{J_0} = Q_1(n)$$

K_0 \ $\overline{Q_0}$ Q_0

$\overline{Q_1}$	X	1
Q_1	X	

$$K_0 = \overline{Q_1}(n)$$



	Q_1	Q_0	
	0	0	0
↑	1	0	2
↑	1	1	3
↑	0	1	1
↑	0	0	

$$Q_1(n+1) = J_1 \bar{Q}_1(n) + \bar{K}_1 Q_1(n)$$

$$= \bar{Q}_0(n) \cdot \bar{Q}_1(n) + \bar{Q}_0(n) Q_1(n)$$

$$Q_1(n+1) = \bar{Q}_0(n)$$

$$Q_0(n+1) = J_0 \bar{Q}_0(n) + \bar{K}_0 Q_0(n) = Q_1(n) \bar{Q}_0(n) + Q_1(n) Q_0(n) = Q_1(n)$$

CLK

$Q_1 Q_0 = 00$

$$Q_0 Q_1 = 00$$

	Q_0	Q_1	
	0	0	0
└─┐	0	1	1
└─┐	1	1	3
└─┐	1	0	2
└─┐	0	0	

- Design counter 0 – 1 – 4 – 3 – 5 using T-FF H.W.

MOD-5

3 ffs

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0						
0	0	1						
0	1	0	X	X	X	X	X	X
0	1	1						
1	0	0						
1	0	1						
1	1	0	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X



Topic : 2 Min Summary

→ Synch. Counter.

Thank you

GW
Soldiers !

